

Understanding Mixed Reality Drift Tolerance

Daniel Killough*

Department of Computer Sciences
University of Wisconsin-Madison
Madison, Wisconsin, USA
dkillough@wisc.edu

Yuhang Zhao

Department of Computer Sciences
University of Wisconsin-Madison
Madison, Wisconsin, USA
yuhang.zhao@cs.wisc.edu

Ruijia Chen*

Department of Computer Sciences
University of Wisconsin-Madison
Madison, Wisconsin, USA
ruijia.chen@wisc.edu

Bilge Mutlu

Department of Computer Sciences
University of Wisconsin-Madison
Madison, Wisconsin, USA
bilge@cs.wisc.edu

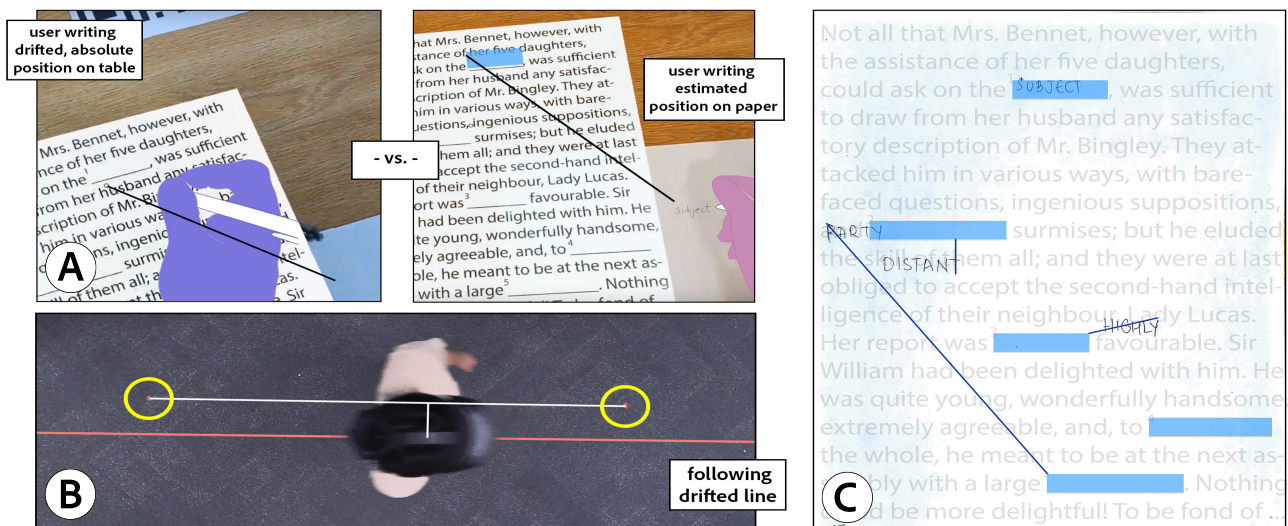


Figure 1: An overview of task interpretations, illustrating: (A) Two primary writing behaviors when responding to drift: following drifted location absolutely versus estimating the un-drifted position; (B) Path deviations from center while following virtual drifted line (red) versus actual invisible line (white); (C) Participant's writing paper with original paper overlaid, showing appropriate blanks (blue boxes) and corresponding written words.

Abstract

Drift is a phenomenon observed in mixed reality (MR) devices caused by inaccuracy in the device's tracking system, leading virtual objects to translate in virtual space away from their intended position. In this paper, we contribute a preliminary study investigating the effects of drift with 10 users, using quantified levels of drift measured by researchers and informed by prior work to measure their performance and perception on a simple task in mixed reality,

seeking to provoke future work and novel conversations around alternate solutions when drift occurs. Our findings demonstrate a need for MR developers to control against severe amounts of drift and provide clear instructions when drift occurs, including qualitative and quantitative "accepted" metrics for drift.

CCS Concepts

• Human-centered computing → Empirical studies in interaction design; Mixed / augmented reality.

Keywords

Interaction Methods, Drift, Augmented and Mixed Reality

ACM Reference Format:

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* Authors contributed equally to this research.

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1 Introduction

Mixed reality (MR) devices interweave virtual objects with real-world environments. Factors like time and sharp movements add “drift” to virtual objects, dragging them astray from their intended positions, impairing users’ experiences [8, 10, 17, 31, 32]. Additionally, MR head-mounted displays (HMDs) may misinterpret the environment, resulting in incorrect recognition of the user’s position [17]. As a result, existing MR HMDs (e.g., Microsoft HoloLens 2) display virtual objects at varying offset from the real-world “anchor”—a physical counterpart of the virtual object or an image target like a QR code [28, 37, 38]. Though prior literature investigates various methods to *combat* drift [11, 12, 15, 20, 41], it lacks a thorough evaluation of users’ *tolerance* of drift. This study aims to fill this gap by examining user tolerance for varying drift levels in a controlled lab environment, thus deriving tolerable drift levels in MR and inspiring MR interface design. In this paper, we evaluate participants’ ability to complete a simple task as the drift amount dynamically changes across five standard offsets: 0cm (baseline), 1.5cm, 5cm, 11cm and 17cm offset added. We seek to answer (RQ1.) How does user performance in an MR task change as the amount of drift increases? and (RQ2.) How does user perception of drift change as the amount of drift increases?

To answer these questions, we developed an MR application on HoloLens 2 asking participants to travel along a virtual path between two physical tables. We presented a fully virtual page (Appendix B) at one table, instructed participants to memorize one word at a time, then walk to the second table to transcribe the word using a physical pen and paper onto a new, virtual page presented in MR (Appendix A). For each word we remotely offset the alignment between the augmented paper and the physical paper, as well as between the virtual path and physical path indicators, to measure how participants follow or account for drift. Through an exploratory study with 10 participants, we found that user performance and perception both significantly worsened at the 11cm level and above compared to the baseline (i.e., 0cm added offset), and users’ interpretation of the writing task varied as the drift increased. Our key contributions include: (1) A preliminary, exploratory study about how users perceive and interpret drift levels in MR; (2) Recommendations for tolerated drift levels should drift occur in regular MR use; and (3) Recommendations for clear task instructions in MR to reduce interpretation ambiguity when drift occurs.

2 Contextualizing MR Terms

We use the term mixed reality (MR) in this work to refer to Microsoft’s definition identified by Speicher et al. [34], but we recognize users may have different definitions of MR versus Augmented Reality (AR). To maintain consistency with prior work we use AR when relevant, but believe “MR” may also be applicable. The *Microsoft HoloLens 2* [23], a popular mixed reality device, uses “Spatial Anchors” to help track the headset’s location in a scene. Spatial Anchors are pre-defined, registered locations in a user’s physical environment, acting as local coordinate systems for developers to attach virtual 3D objects (“holograms”) to [28, 37]. Microsoft defines

the “quality” of holograms by describing how holograms appear relative to their physical counterparts, notably the perceived “accuracy” (whether a hologram appears at its originally locked location) and “drift” (how a hologram moves away from its originally anchored location) [38]. In this work we use these systems to evaluate our users’ perception of drift in static (writing task) and dynamic (walking task) contexts, particularly how their experience falters when Spatial Anchors fail to properly identify their surroundings.

3 Related Work

3.1 Evaluating Existing Tracking Systems

Existing evaluations of the HoloLens’ tracking method includes Hübner et al.’s work, noting the device’s high degree of precision ($\mu = 0.2cm$) to calibrate location in space using its built-in depth sensor and tracking systems [15]. Furthermore, work by Guinet et al. found that HoloLens tracking is pretty close to motion capture (their hard-truth baseline) [12]; however, drops in performance can affect the likelihood and perceived impact of drift, leading to a potentially significantly worse experience from the user’s perspective. Although Microsoft provides some initial recommends for running apps at a consistent framerate of 60 frames-per-second (FPS) to match virtual and physical objects as closely as possible from the user’s perspective [13, 38], these systems need alternate solutions when existing systems fail.

One such solution on Spatial Anchors exists in Microsoft’s World Locking Tools [20], which have demonstrated less drift than individual anchors across a scene [11, 29] as points in space appear “fixed” to one another instead of being allowed to dynamically relocate themselves as the headset moves. Some existing systems expand further upon the accuracy and drift of holograms from the World Locking Toolkit, like *Nail It!* [11], which corrects for mixed reality drift by taking a series of images with the headset, comparing them to “hard-truth”, pre-taken keyframes, and correcting accordingly. Some research explored bundle adjustment [39, 42] and monocular visual-inertial state estimation [18] to reduce drift and improve the quality of the reconstruction.

Other studies also use MR head-mounted displays to map [15] or navigate [12, 41] environments, but none evaluate perception or performance of recalibrating headset orientation by their end-users. Given that human capabilities can be altered by the tools they use, ensuring alignment between virtual objects and the physical world is crucial. Considering the wide variety of interface scales across different tools, it’s imperative to study how users tolerate drift to ensure the continuity and accuracy of user experiences and interactions in MR environments.

3.2 Perception of MR Drift Inconsistencies

3.2.1 Self-Location. One of the theoretical concepts behind drift perception is the theory of self-location [6], particularly important in virtual and augmented reality research to help users orient themselves in a 3D scene [25]. Stability of head movement has been shown to improve perception of self-location [16]. Likewise, studies have compared perceived self-location in regards to motor versus perceptual tasks [25]. Yet, within perceptual tasks, there still lacks measurement of self-location regarding mixed reality drift correction methods. Drift is often considered an important factor in

regards to a user’s sense of self-location [31, 32]. Furthermore, prior work has shown that reducing drift also improves self-location [17] and reduces nausea [30]. Prior work has explored methods in which people attempt to physically counteract the effects of drift [8, 10]. Although this work builds upon prior work that declares how self-location improves with stability of objects in the environment [35], there lacks an exploration of the various means of software correction to drift in regards to improving self-location.

3.2.2 Registration Error. A key issue in MR applications is the inconsistency between virtual information and the real environment [5], also called “registration error” [14]. From the user’s perspective, registration errors can disrupt users’ experience and the sense of realism in mixed reality, breaking the illusion that real and virtual objects co-exist [24]. Despite its significance, research quantifying the effects of registration errors in mixed reality remains limited. For example, Wang et al. [40] examined human sensitivity to changes in the size of virtual objects within augmented reality (AR), specifically focusing on the detection threshold for these changes. Li et al. [19] explored the perception of body ownership in AR, finding that the noticeability of movement inconsistency increases approximately quadratically as the offset size enlarges. Terrier et al. [36] studied the impact of registration errors on user behavior during a pick-and-place task, observing that while users maintained consistent behavior, any registration errors that hindered task completion led to noticeable disruptions. However, their study was conducted in a simulated AR environment in virtual reality (VR). In this paper, we aim to further quantify drift tolerance in a real-world MR environment and explore how task performance, perception of the MR environment, and task interpretation change as drift increases.

4 Method

4.1 Participants

We recruited a convenience sample of 10 UW-Madison students. Participants varied in age from 18 to 28 (mean = 23.4, SD = 3.2), gender (9 male, 1 female). 9 of 10 participants had prior AR or MR experience (excluding P4), including games (e.g., *Pokémon Go* [26], *Nintendo 3DS AR Cards* [27]), filters (e.g., *Snapchat* [33]), headsets (e.g., *Hololens* [23], *Apple Vision Pro* [3]), and tools (e.g., *Amazon AR View* [1]). Condition ordering was distributed to the participants using latin square counterbalancing to standardize condition exposure. We repeated the same 5x5 latin square twice. Demographic information for each participant can be found in Table 1. This study was approved by the UW-Madison Institutional Review Board (IRB).

4.2 Study Design

To investigate the effects of drift in MR, we chose two simple, mutually exclusive tasks to address dynamic locomotion (e.g., Guinet et al.’s walking trial [12]) and static precision components found in MR apps (e.g., Andrews et al.’s clinical task with attached tracking marker [2]). We asked participants to walk between two tables (locomotive, dynamic component) and write on a virtual sheet of paper (precision, static component).

Table 1: Participant demographics and latin square order.

PID	Age	Gender	Past AR/MR Exp?	Drift Amount	Order Shown
P1	19	Male	AR games, filters	1.5cm, 5cm, 11cm, 17cm, 0cm	
P2	18	Male	AR games, filters	5cm, 11cm, 17cm, 0cm, 1.5cm	
P3	24	Male	MR headset	11cm, 17cm, 0cm, 1.5cm, 5cm	
P4	28	Male	None	17cm, 0cm, 1.5cm, 5cm, 11cm	
P5	26	Male	MR headset	0cm, 1.5cm, 5cm, 11cm, 17cm	
P6	24	Female	MR headset	1.5cm, 5cm, 11cm, 17cm, 0cm	
P7	27	Male	AR games	0cm, 1.5cm, 5cm, 11cm, 17cm	
P8	22	Male	AR filters, tools	11cm, 17cm, 0cm, 1.5cm, 5cm	
P9	23	Male	AR games	17cm, 0cm, 1.5cm, 5cm, 11cm	
P10	23	Male	MR headset	5cm, 11cm, 17cm, 0cm, 1.5cm	

Environmental Setup. To ensure minimal drift added by the headset, we conducted a single-session study in a controlled, well-lit laboratory environment with windows covered. We set up two tables distanced approximately 10 meters from each other: One table holds a blank sheet of 8.5x11 standard US letter paper (the “info table”) and a set of QR codes. The other table holds a blank sheet of identical paper, a set of QR codes, and a pen (the “writing table”). Between the tables, we placed four tape marks on the ground spaced symmetrically 3m apart in a straight line to indicate the path. Each table’s paper virtually showed a same excerpt from *Pride and Prejudice* [4], but on the writing table, five words were missing (Appendix A); on the info table, the same five words were highlighted (Appendix B).

Procedure. Between the tables, the headset displayed a solid-colored, highlighted path that either aligned with or deviated from the tape marks on the floor. Participants began at the writing table, traveled along the highlighted path to the info table, and learned the word needed to fill in the blank. Each time a participant left the writing table, the researcher adjusted the drift levels of the virtual paper and path for one of five experimental trials, using random assignment and Latin square counterbalancing to determine the trial order. After reaching the info table, participants memorized the word, returned to the writing table, and filled it in on the drifted form. This process was repeated five times, once for each drift level, with participants providing feedback about their experience after each trial through a series of Likert questions outlined in Appendix D. We provide an illustration of the study setup in Figure 2:

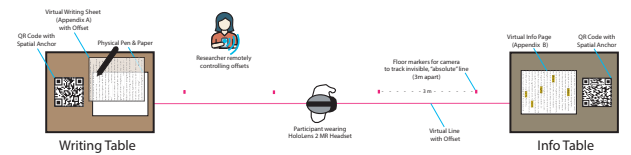


Figure 2: Study Setup

To ensure accurate measurement of different drift conditions, we conducted a preliminary examination to assess the device’s inherent drift under different conditions. Utilizing Microsoft’s Spatial Awareness system, we determined that the average drift amounted to approximately 1.5 centimeters, consistent with findings reported in previous studies (1.9 ± 0.4 cm and 1.3 ± 0.1 cm) [15]. We also

conducted trials without the spatial awareness system, where drift measurements from three separate tests yielded values of 0cm, 5cm, and 17cm. Based on these measurements, we established five conditions for our independent variable, manually offsetting digital objects by their original locations to represent varying extents of drift: A control condition representing 0cm (no additional offset added), 1.5cm offset added (average drift with Microsoft's Spatial Awareness system), 5cm offset added (the minimum non-zero drift condition identified without the spatial awareness system), and 17cm offset added (the maximum non-zero drift condition identified). We added an additional condition of 11cm offset added as the midpoint non-zero drift condition between 5cm and 17cm. After standardizing the offset amounts we randomly calculated direction vectors for each level, enumerated in Table 5 (Appendix C). We offset the writing sheet by each drift condition magnitude (0cm–17cm) using randomly generated unit vectors, keeping the y-axis unchanged to prevent it from appearing under the physical table. Since the walking path had functionally infinite length between the two tables, we offset the path either left or right against each table's z-axis. We did not change values between participants.

Apparatus. We built the experiment platform on Microsoft HoloLens 2, with Unity 2021.3.31f1 and Microsoft Mixed Reality Toolkit 3 [21]. We placed QR codes [22] on both the info and writing tables to recalibrate the headset twice during each trial, ensuring that the offsets applied were aligned with the predetermined values. We sent the intended offset amounts wirelessly to the headset using a synchronized smartphone application built with React Native. To record path deviation, we mounted an Insta360 X3 360° video camera directly above the midpoint of the path to capture video and audio. Additional audio sources for pre- and post-task interviews were recorded by the researchers' cell phones and tablets.

4.3 Data Collection

During the study, researchers measured: (1) Locations participants filled in each blank on the form; (2) Distances of deviations from the physical path; and (3) User perceptions of offsets impacting the task. Rationalized by prior work on self-location (§ 3.2.1) we hypothesize that drift will (H1) reduce participant performance and (H2) reduce user perception if drift reduces a sense of self-location.

4.4 Analysis

For each set of quantitative data, we checked the normality of each measure using the Shapiro-Wilk test for normality. None of these measures were normally distributed, so we used the Aligned Rank Transform (ART) ANOVA [9] to evaluate the effect of different measures, including where each participant wrote from the intended location, deviations from the virtual path, and responses to our likert questions. If statistically significant, we used the ART contrast test for post-hoc comparison. We used partial eta squared (η_p^2) to calculate the effect size, with 0.01, 0.06, 0.14 representing the thresholds of small, medium and large effects [7].

For qualitative analysis, we audio recorded all studies and transcribed interviews using an automatic transcription service. We analyzed the transcripts using thematic analysis—two researchers open coded two identical samples (20% of the data) independently, and they developed an initial codebook by discussing their codes to

resolve any disagreements. The two researchers then separated the remaining transcripts and coded independently based on the codebook. During this process, they periodically checked each other's work and discussed their codes to ensure consistency. After the researchers reached an agreement, new codes were added to the codebook. We then developed themes and sub-themes, clustering relevant codes using axial coding and affinity diagrams. Once initial themes and sub-themes were identified, researchers cross-referenced the original data, the codebook, and the themes to make final adjustments, ensuring all codes were correctly categorized.

5 Results

5.1 Drift Causes Fundamental Interpretation Differences

During our study, we found that two interpretations emerged from our instructions: “Fill in the corresponding blank on the form at the writing table”: (1) Participants wrote words based on their absolute location displayed in augmented reality, or: (2) Participants attempted to position words according to the offset amount and wrote them at the estimated location. In our study, eight participants chose Interpretation 1 (P1–P7, P10), while two participants chose Interpretation 2 (P8 and P9). Within Interpretation 1, three participants wrote words totally off-page, with these participants choosing to write on the table when the offset was too large (P3, P4, P5)—twice in the 11cm condition (P4, P5) and twice in the 17cm condition (P3, P4). Two of these participants also wrote words partially off-page, once in the 5cm condition (P4) and once in 11cm (P3). Other participants had manners of managing drift, like cramming words tightly on the page margin (e.g., P1, P2) as close to the off-page blank as possible, or P7 writing “crown ->” with an arrow pointing off-page.

As for the offsets' usability, many participants agreed that offsets affected their task completion and made the task confusing (P1–P8); P2 even wrote a word in the wrong box. The threshold for being unusable was determined to be between 5cm and 11cm—Most participants agreed that 1.5cm felt manageable (P1, P3, P5, P7, P9, P10), but 5cm was the threshold for feeling worse (P3, P7). Some participants explicitly considered offsets 11cm and larger unusable (P1, P2, P7, P9), while some participants started from 5cm (P3, P5, P6, P8). When asked to rank amounts of offset from ‘best’ to ‘worst’, participants generally responded in order with 0cm best and 17cm worst. However, some participants considered the task difficulty under 0 and 1.5cm offset as equally easy (P1, P4, P6, P10). When the offset made the blank partially or totally off-page, multiple participants preferred a larger offset if the word was entirely on the page (P1, P3, P5, P6, P7). For example, P9 considered the 17cm offset as “doable but pretty bad”, but when 11cm's offset made the word completely off-page, it was “not usable ... horrible”. Additionally, P3 thought any offset causing a word to be half-off page was more confusing than entirely off-page.

We found a significant main effect of offset amounts on the deviation distances of words ($F_{4,32} = 22.756, p < .001, \eta_p^2 = 0.74$). Additionally, there was a significant main effect based on the interpretation ($F_{1,8} = 25.092, p = .001, \eta_p^2 = 0.76$) on deviation distances of words. Furthermore, the interaction between the amounts of offset and the interpretation was significant ($F_{4,32} = 16.8944,$

$p < .001$, $\eta_p^2 = 0.68$). Post-hoc analyses revealed a significant difference in mean deviation distances of words between offset amounts 0cm and 1.5cm ($t_{ratio} = -3.499$, $p = .010$), 0cm and 5cm ($t_{ratio} = -5.729$, $p < .001$), 0cm and 11cm ($t_{ratio} = -6.999$, $p < .001$), 0cm and 17cm ($t_{ratio} = -10.405$, $p < .001$), 1.5cm and 11cm ($t_{ratio} = -3.499$, $p = .010$), 1.5cm and 17cm ($t_{ratio} = -6.906$, $p < .001$), 5cm and 17cm ($t_{ratio} = -4.676$, $p < .001$), and 11cm and 17cm ($t_{ratio} = -3.406$, $p = .013$). Full results from participants' word distances can be found in Table 2. These results showed that both the offset amount level and how people interpreted the task affected their performance in words' deviation distances, and these two factors intersect.

Table 2: Word Distances from Baseline (cm) organized by first condition shown. Particularly notable cells include P2's 0cm condition written in the wrong box; values with an asterisk (*) (P3, P4) were written partially off-page and on the table; values with a cross (†) (P3, P4, P5) were written fully on the table and recorded from where the user started writing.

Drift	0cm first		1.5cm first		5cm first		11cm first		17cm first	
	P1	P6	P2	P10	P3	P8	P4	P9	P5	P7
0cm	0.37	0	8.1	0.38	0.08	0	0	0	0.11	0
1.5cm	0.84	1.74	2.15	1.47	0.41	1.11	1.51	1.28	1.8	0.98
5cm	3.38	1.85	4.52	6.45	1.58	1.22	5.36*	0.14	3.2	2.57
11cm	9.6	8.15	8.87	10.8	10.95*	0.76	11.81†	0.38	12.29†	0.83
17cm	16.36	16.44	15.47	15.76	16.66†	1.54	17.04†	1.84	6.5	16.66

5.2 Drifted Virtual Path

From our video recordings, we calculated the distances by which participants deviated from the actual route markers under different offset conditions. Offset direction (left or right) was randomized during the experiments and can be found in Table 5. An illustration of how the distances were calculated can be found as letter B in the teaser figure 1. While we did find a significant effect of offset amounts on the deviation distances of the path ($F_{4,36} = 2.866$, $p = 0.037$, $\eta_p^2 = 0.24$), no significant differences were found in post-hoc analysis. Therefore, we did find that adding offsets impacted how far users deviated from the path, but no condition differed from another at specific thresholds.

Table 3: Average Path Deviations from Center (cm)

Drift	0cm first		1.5cm first		5cm first		11cm first		17cm first	
	P1	P6	P2	P10	P3	P8	P4	P9	P5	P7
0cm	4.06	7.08	9.1	0.58	9.33	13.65	5.51	5.83	8.52	5.26
1.5cm	6.4	7.53	6.54	6.77	4.93	14.15	5.33	4.63	5.81	6.8
5cm	3.65	4.8	5.05	6.9	11.04	17	2.85	7.97	2.81	6.08
11cm	16	8.3	9.95	16.16	6.95	6.07	10.95	7.05	14.94	9.48
17cm	6.46	11.2	10.89	18.69	19.38	4.69	11.1	1.38	9.12	12.75

5.3 Significant differences in task ease across offset amounts

After each trial, we asked participants to rate “How easy did this offset make it to complete the task?” on a Likert scale, where 1

stands for most difficult and 7 stands for most easy. We found that offset amounts significantly impacted participants' perception of task difficulty ($F_{4,36} = 18.295$, $p < .001$, $\eta_p^2 = 0.67$). Post-hoc analysis revealed a significant difference in mean Likert scale scores between offset amounts 0cm and 5cm ($t_{ratio} = 4.102$, $p = .002$), 0cm and 11cm ($t_{ratio} = 7.038$, $p < .001$), 0cm and 17cm ($t_{ratio} = 6.4095$, $p < .001$), 1.5cm and 11cm ($t_{ratio} = 5.422$, $p < .001$), 1.5cm and 17cm ($t_{ratio} = 4.794$, $p < .001$), and 5cm and 11cm ($t_{ratio} = 2.936$, $p = .043$). These results suggest that varying offset levels significantly influenced participants' perception of task difficulty. Specifically, significant differences were found between the smaller offsets (0cm and 1.5cm) and the larger offsets (11cm and 17cm).

This significant variation in perceived difficulty between offset levels is further supported by participants' qualitative feedback. We asked participants to rate their agreement with the statement, “To me, there was no significant difference between each of the offsets.” on a likert scale, with 1 standing for strongly disagree and 7 for strongly agree. Most participants thought offsets made the task difficult (mean = 2.7, SD = 1.95), and only 2 participants gave scores higher than 4 (neutral). The two participants who gave scores higher than 4 noted the overall ease of the task (P9) and only difficult if the offset caused the blank to exit the page (P10), as that added “physical danger” that impacted P10's ability to write: “Because I had to translate my actual writing.”

5.3.1 Offset Impacting Immersion. To measure how drift affected user immersion, we also asked participants to rate, “I felt immersed throughout the experience.” on a Likert scale, where 1 stands for strongly disagree and 7 stands for strongly agree. Most participants are neutral on this statement (mean = 4.1, SD = 1.73). We found that participants considered alignment as an important component of immersion, and offsets disrupted them from a seamless experience. Five participants mentioned that since the virtual paper was mostly opaque, they could not see their handwriting well. This behavior meant that the virtual paper occluded the writing paper, causing P1, P7, and P8 to need to tilt their heads to see the real page well, harming their immersion. However, we did not find that the virtual paper's opacity significantly impacted participants' perception of different drift levels, and participants reported the task was still manageable (P1, P6, P10).

Table 4: Likert ratings for each participant. Half point values were recorded for participants who provided two ratings.

PID	0 cm	1.5 cm	5 cm	11 cm	17 cm	“No Sig. Difference”	“I Felt Immersed”
P1	7	7	4.5	2	4	Disagree (2)	Slightly Disagree (3)
P2	7	4	7	6	3	Disagree (2)	Slightly Disagree (3)
P3	6	6	5	3	4	Strongly Disagree (1)	Strongly Agree (7)
P4	7	7	4	4	3	Strongly Disagree (1)	Slightly Agree (5)
P5	6	6	5	1	2	Strongly Disagree (1)	Neutral (4)
P6	5	4	3	3	3	Neutral (4)	Neutral (4)
P7	7	7	6	1	4	Disagree (2)	Agree (6)
P8	7	5	3	2.5	2	Disagree (2)	Strongly Disagree (1)
P9	7	6	4	3	6	Agree (6)	Slightly Agree (5)
P10	7	7	6	6	5	Agree (6)	Slightly Disagree (3)

6 Discussion

6.1 Usable drift threshold

Users seem to have a certain tolerance for drift in mixed reality systems. While we did measure a steady decrease of task performance across multiple measures as drift increased, all subjects considered the smallest drift amount of 1.5cm to be usable. This suggests that hyper-accurate drift correction might not have to be a high priority for mixed reality device manufacturers in certain cases.

6.1.1 Different performances for intrinsic and extrinsic tasks. Curiously, drift had varying effects based on the task. Performance on the writing task was highly influenced by our offsets, but performance on the line-following task showed no significant difference in post-hoc analysis between any two offset amounts. We believe the line-following task's invariance to drift was likely because 1) the line was perfectly straight and 2) the destination was clearly visible throughout the user's entire journey along the line. In other words, the line was merely a suggestion; as long as the user understands the line directs them between the two tables, they do not necessarily need the line when performing the task. This finding suggests that the size of the landmark labeled by a virtual object is important to drift tolerance in the task— Since the table is large, the precise location of the label (the line in this case) is non-essential assuming it lands somewhere near the table, and therefore MR drift can be adapted to without consequence. However, the writing task, existing within the context of its corresponding physical objects, established a spatial relationship dependent on the pen and paper. This relationship may have caused more confusion and led to task ambiguity with participants. These results highlight that drift tolerance is closely related to the task's intrinsic nature and how important virtual holograms are to task execution in MR environments.

6.1.2 Differences in Interpretation. Two users attempted to correct for the perceived drift themselves by writing the words where it appeared the underline should be if the drift were corrected (Interpretation (2) §5.1). These users were negatively impacted by drift, but they were able to write the words closer to the true location of the corresponding underline, with the trade-off likely being that it took them longer to complete the task. This suggests that if the task is not time-sensitive, users may be able to work around large drift amounts. However, since only two out of ten users did this interpretation unprompted, it would be wise for application designers to suggest this strategy to users if they suspect their app is prone to drift.

6.1.3 Drift negatively impacts user immersion. All users stated there was at least some perceivable difference between the offset amounts, with most agreeing that the difference was very noticeable. There was also a significant increase in perceived task difficulty for all offset levels compared to zero. Furthermore, users considered higher amounts as less immersive when performing the tasks. We find that, while small drift amounts can have minimal impacts on task performance, users still notice drift and feel that it negatively impacts their immersion.

6.2 Implications for Future Research and Design

Future work should consider tracking participant performance and cognitive load, such as the time taken to complete each task and the amount of concentration required to complete the task during each of the two interpretations. With users rating drift levels of 11cm as “unusable” (P9), we classify 11cm and 17cm as “severe” drift levels that should be controlled against in all mixed reality applications where virtual objects are intended to align with real-world counterparts. Furthermore, we also recommend mitigating against any drift levels over 5cm even if tolerated by some users.

6.3 Limitations

Limitations of our study include the inability to precisely track the position of participants over time in the study. Precision could be added using an OptiTrack or other high-performance motion capture system. Our simple task system should be considered as one example of a mixed-reality aided task; future work should also explore tasks in real-world task areas like manufacturing, with varying amounts of pre-training, instructions, and intrinsic & extrinsic factors. Additionally, our work primarily focuses on participants' perception of different drift levels and does not evaluate metrics such as task completion time and word-fill correction rate. Future work should incorporate these additional measurements to gain deeper insights and use standardized questionnaires, such as the NASA-TLX, for more comprehensive evaluation. Furthermore, although our drift levels are founded on measured amounts and prior work, these levels are discrete steps that do not consider intermediate drift levels. Based on our results, future work may also analyze between the 5cm and 11cm conditions and perhaps the 1.5cm and 5cm conditions to more specifically target tolerated maximum drift values on various tasks. The robustness of our findings would also be strengthened with more participants of more varying demographics (e.g. above the age of 28, more female participants, more participants with no prior augmented or mixed reality experience).

7 Conclusion

In this paper, we investigated the effects of drift in mixed reality devices on user task performance and subjective experience. We had 10 study participants complete a mixed reality activity under five discrete levels of drift and recorded how correctly they performed the tasks alongside their perceptions of the experience during and after the activity. We found that a small amount of drift is tolerable for most users with respect to both their performance and their perception. However, their opinion degrades faster than their actual task performance as digital objects are offset further from their original location. When faced with drift, study participants managed in one of two ways: Most attempted to complete the task as if the drift were nonexistent, while a minority attempted to manually correct for the drift. These results suggest that mixed reality app developers can permit a small amount of drift while not sacrificing user immersion. Such drift tolerance may be lesser or greater depending on the elements associated with the task, and how precise users need to be to perform the task adequately. We also advise developers to instruct users on what to do if they encounter drift to avoid unpredictable user behavior, as some may attempt to correct for drift while others may follow exactly as the headset advises.

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A Writing Sheet

Not all that Mrs. Bennet, however, with the assistance of her five daughters, could ask on the ¹_____, was sufficient to draw from her husband any satisfactory description of Mr. Bingley. They attacked him in various ways, with barefaced questions, ingenious suppositions, and ²_____ surmises; but he eluded the skill of them all; and they were at last obliged to accept the second-hand intelligence of their neighbour, Lady Lucas. Her report was ³_____ favourable. Sir William had been delighted with him. He was quite young, wonderfully handsome, extremely agreeable, and, to ⁴_____ the whole, he meant to be at the next assembly with a large ⁵_____. Nothing could be more delightful! To be fond of ...

B Info Sheet

Not all that Mrs. Bennet, however, with the assistance of her five daughters, could ask on the ¹**subject**, was sufficient to draw from her husband any satisfactory description of Mr. Bingley. They attacked him in various ways, with barefaced questions, ingenious suppositions, and ²**distant** surmises; but he eluded the skill of them all; and they were at last obliged to accept the second-hand intelligence of their neighbour, Lady Lucas. Her report was ³**highly** favourable. Sir William had been delighted with him. He was quite young, wonderfully handsome, extremely agreeable, and, to ⁴**crown** the whole, he meant to be at the next assembly with a large ⁵**party**. Nothing could be more delightful! To be fond of ...

C Drift Direction Vectors

Table 5: Randomly generated drift unit vectors for each offset amount including the X,Z components of each unit vector applied to the writing sheet. The Info Dir and Writing Dir columns show directions based on the Z axis of that table.

Drift	Writing Sheet Vector (x,z)	Info Dir	Writing Dir
0cm	0.0f, 0.0f	None	None
1.5cm	-0.99548f, -0.09495f	Right	Left
5cm	0.28123f, -0.95964f	Left	Left
11cm	0.01256f, -0.99992f	Right	Right
17cm	0.77652f, 0.63009f	Left	Right

D Likert Survey Questions

D.1 Pre-Task Interview Questions:

- (1) Age
- (2) Gender
- (3) To your knowledge, have you had any experience with AR technologies? (e.g. Pokemon Go, Snapchat Filters)

D.2 Between each task condition:

D.2.1 Likert scale, 1 = most difficult; 7 = most easy.

- (1) How easy did this offset make it to complete the task? Why?

D.3 Post-Task Interview Questions:

D.3.1 Likert scale, 1 = strongly disagree; 7 = strongly agree.

- (1) "To me, there was no significant difference between each of the offsets." Why?
- (2) "I felt immersed throughout the experience." Why?
- (3) *If they didn't feel immersed, ask:* At what point were you no longer immersed in the scene?

D.3.2 Qualitative Questions.

- (1) How did the offset amounts impact your ability to complete the task?
- (2) What challenges did you encounter while using this headset?
- (3) Rate each amount of offset from 'best' to 'worst'. Why did you choose this order?
- (4) Were all of these offset levels usable? Why or why not?
- (5) *If not all offset levels were usable, ask:* At what point did it stop becoming usable?